Circles-1

Important Properties :

P-1 ⇒ Equal chords subtend equal angles at centre & vice-versa.



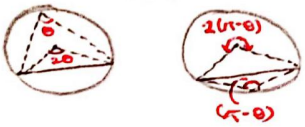
P-2 ⇒ Equal chords are equidistant from centre.

P-3 ⇒



Arc subtends double angle at centre | Angle at all pts is same | Diameter subtends 90°

Centre pe double angle subtend hoga.



Angles in Major & Minor Segment.

P-4 ⇒ Angles in same segment are equal.



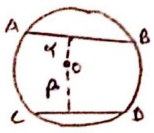
Angles in minor segment | Angles in major segment

P-5 ⇒ Product of divisions of two intersecting chords is equal.



$AP \cdot BP = CP \cdot DP$ | $AB \cdot BP = CD \cdot DP$

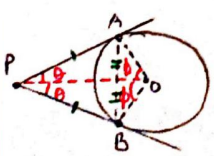
P-6 ⇒ Longer the chord, nearer it is to centre.



$AB > CD \Rightarrow \alpha < \beta$] Therefore, diameter is longest chord

P-7 ⇒ Radius is always ⊥ar to tangent at pt. of contact. Hence all normals pass through centre.

P-8 ⇒ Tangents from an ext. pt. are equal.



$PA = PB$

→ PAOB is a cyclic quad.

→ Sum of opp. angles $= \pi$

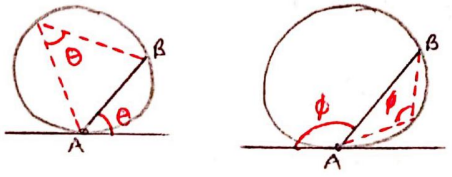
→ Line joining P & O behaves as angle bisector.

$\angle PAO + \angle PBO = \pi$

$\angle APB + \angle AOB = \pi$

P-9 (Alternate Segment Theorem) ⇒

If a line touches a circle & chord is drawn from that point, then angle which chord makes with line is equal to the angle made by chord in corresponding alternate segment.

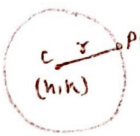


P-10 ⇒ Derived from P-5 (sp. case)



$PA \cdot PB = PT^2$

General Eqn of Circle :



$(x-h)^2 + (y-k)^2 = r^2$

Also,

$x^2 + y^2 + 2gx + 2fy + c = 0$

Here, Centre $\equiv (h,k) \equiv (-g,-f)$

$r = \sqrt{g^2+f^2-c}$

Cond^n for 2nd degree general Eq^n to represent circle ⇒

1) Coeff. of x^2 = Coeff. of y^2
2) Coeff. of xy $(h) = 0$
3) Three arbitrary ~~cons~~ constants (g, f & c) are present in eqn.

↓

Therfore a unique circle always passes three non-collinear pts (3).

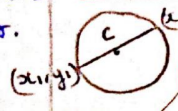
Nature of Circle ⇒

If $g^2+f^2-c > 0$ ⇒ Real circle with finite radius

$g^2+f^2-c = 0$ ⇒ Point circle (Real)

$g^2+f^2-c < 0$ ⇒ Imaginary Circle

Diametric form of Circle :



$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$

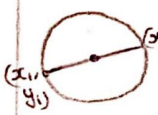
→ Where (x_1,y_1) & (x_2,y_2) are the diametric ends of the diameter.

→ Above expression is also the sum of two quadratics in x & y, whose roots are $(x_1 \& x_2)$ & $(y_1 \& y_2)$ respectively and leading coeff is 1.

$$x^2 - (x_1+x_2)x + x_1x_2 = 0$$

$$y^2 - (y_1+y_2)y + y_1y_2 = 0$$

→ Hence, we can get eqn of circle by simply adding two quad. after making leading coeff 1.



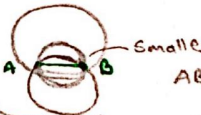
$ax^2 + bx + c = 0$ ⟨ x_1, x_2

$py^2 + qy + r = 0$ ⟨ y_1, y_2

$$\Rightarrow \left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) + \left(y^2 + \frac{q}{p}y + \frac{r}{p}\right) = 0$$

↳ Eqn of given Circle

NOTE : Circle of least radius through 2 pts is the one formed by taking those pts as diametric ends.

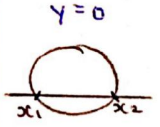


Smallest when AB = diameter

Intercepts of a Circle :

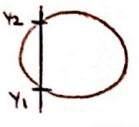
→ Intercept made by the circle is the distance b/w two points where circle cuts the line / axis.

For x-int. put $y = 0$



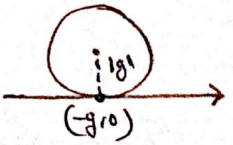
$|x_1 - x_2| = 2\sqrt{g^2-c}$

For y-int. put $x = 0$

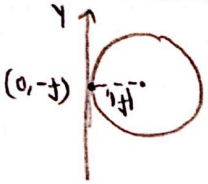


$|Y_1 - Y_2| = 2\sqrt{f^2-c}$

→ If $g^2 - c^2 = 0$ then circle touches x-axis at $x_1 = -g$ and $r = |g|$



→ Similarly, if $f^2 - c = 0$ then circle touches y-axis at $y_1 = -f$ and $r = |f|$



Important Points ⇒

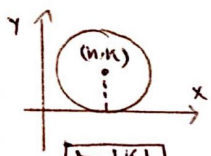
(i) Centre of circle touching both axes would always lie on the line $y = x$ or $y = -x$ and eqn of circle is

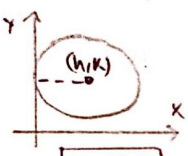
$(x-h)^2 + (y-k)^2 = r^2$
OR
$(x-h)^2 + (y+k)^2 = r^2$





(ii) If circle touches only x-axis or y-axis, then its radius would be y-coordinate or x-coordinate of centre respectively





Position of a point w.r.t circle :

→ Put the coordinates of point in eqn of circle and calculate Power of point (S_1).

$S_1 : x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

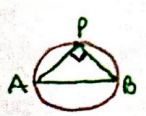


If,

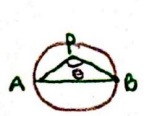
Position of P (x_1, y_1)

- $S_1 < 0$ ⇒ Inside circle
- $S_1 = 0$ ⇒ On periphery of circle
- $S_1 > 0$ ⇒ Outside circle

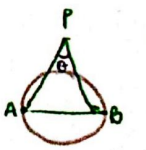
→ Position of a point can also be determined by angle made by diameter at point P.



$\theta = 90°$
⇒ On circle



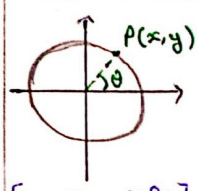
$\theta > 90°$
⇒ Inside circle



$\theta < 90°$
⇒ Outside circle

Parametric Form of Circle :

→ For circle centered at origin ⇒



$C : x^2 + y^2 = a^2$

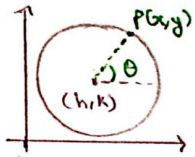
Then,

$x = a\cos\theta$

$y = a\sin\theta$

[θ = Eccentric angle of P]

→ For circle centered at (h,k) ⇒



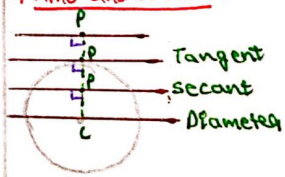
$C : (x-h)^2 + (y-k)^2 = a^2$

$x = h + a\cos\theta$

$y = k + a\sin\theta$

[θ = Eccentric angle of P]

Line and Circle :



$S : x^2 + y^2 = r^2$

$L : y = mx + c$

P : Dist from centre on line

$p = \dfrac{|c|}{\sqrt{1+m^2}}$ (length of ⊥ dist from centre)

If, $p > r$ ⇒ Line passes outside the circle
$p = r$ ⇒ Line is tangent to circle
$p < r$ ⇒ Line is secant to circle
$p = 0$ ⇒ Line is diameter of circle

Alternatively ⇒

Solve the eqn of line & circle by $x^2 + (mx+c)^2 = r^2$

$D > 0$ = Line is a secant
$D = 0$ = Line is a Tangent
$D < 0$ = Line passes outside the circle

Tangent to circle :

(I) Cartesian Form ⇒

Cartesian eqn of tangent drawn to any second degree curve at $P(x_1, y_1)$ can be obtained directly by substituting Variable as following.

$x^2 = x x_1$ $\quad 2y = y + y_1$

$y^2 = y y_1$ $\quad 2xy = x y_1 + y x_1$

$2x = x + x_1$

→ Tangent drawn to any arbitrary circle ⇒



$S : x^2 + y^2 + 2gx + 2fy + c = 0$

$T : xx_1 + yy_1 + g(x+x_1) + f(y+y_1) = 0$

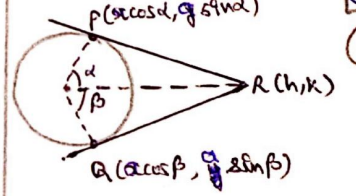
→ Tangent drawn to circle centered at origin ⇒



$S : x^2 + y^2 = a^2$

$T : xx_1 + yy_1 = a^2$

(II) Parametric Form ⇒



$S : x^2 + y^2 = a^2$ (Circle is centered at origin)

$T : xx_1 + yy_1 = a^2$

⇒ $T = x\cos\alpha + y\sin\alpha = a$

→ To calculate P.O.I of a pair of tangent on the circle $x^2 + y^2 = a^2$, we use

$$h = a\left[\frac{\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}\right] \qquad K = a\left[\frac{\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}\right]$$

(Denominator same hai dono ka)

(III) Slope Form ⇒

→ For circle centered at origin ⇒

→ Line $y = mx + c$ would be tangent on $x^2 + y^2 = a^2$ only if

$c^2 = a^2(1+m^2)$ Cond^n of Tangency

(Applicable for a general circle too)

→ It gives us two unique & parallel tangents of m-slope.

$y = mx \pm a\sqrt{1+m^2}$



→ Both the tangents would be diametrically opposite.

→ Point of contact would be diametric ends

Circles-2

→ For a general circle

① Cond^n of tangency is same,

$$c = a\sqrt{1+m^2}$$

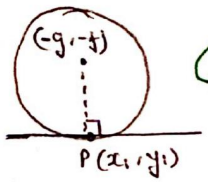
② Replace $x \to (x-h)$ & $(y \to y-k)$

$$T: (y-k) = m(x-h) \pm a\sqrt{1+m^2}$$

↳ Two unique & parallel tangents
↳ Diametrically opposite

Normal to the Circle:

→ Normal to a point in circle is always ⊥ to tangent & passes through centre of the circle.



$S: x^2+y^2+2gx+2fy+c=0$

$$N: y-y_1 = \frac{y_1+f}{x_1+g}(x-x_1)$$

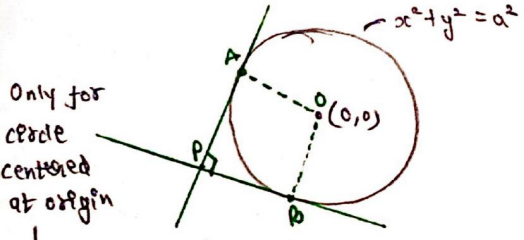
Also,

$$m_N \cdot m_T = -1$$

$m_N \to$ Slope of normal
$m_T \to$ Slope of tangent

Director Circle:

→ It is locus of a point 'P' which moves in such a way that the pair of tangents drawn from P are always ⊥ to each other.

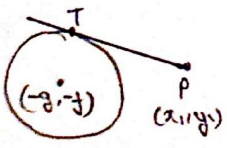


Only for circle centered at origin

Director Circle: $x^2+y^2 = 2a^2$

↳ It itself is a concentric circle of $\sqrt{2}a$ radius

Length of Tangent:



$$PT = \sqrt{S_1}$$

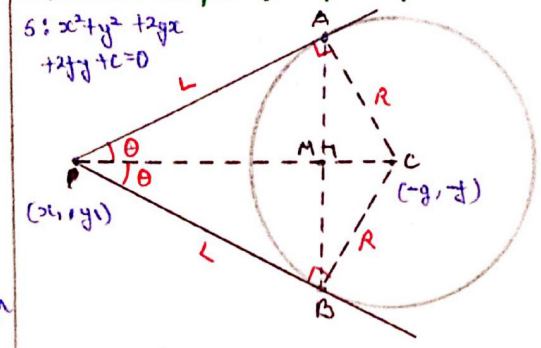
$$PT = x_1^2+y_1^2+2gx_1+2fy_1+c$$

NOTE: To apply this $PT = \sqrt{S_1}$ formula, coeff. of x^2 & y^2 must be 1.

V.Imp

Application of length of tangent ⇒

$S: x^2+y^2+2gx+2fy+c=0$



(i) ar(PACB)

Sol^n $ar(PAC) = \frac{1}{2} \times L \times R$

$$ar(PACB) = ar(PAC) \times 2 = LR$$

(ii) Length of AB

Sol^n $AM = L\sin\theta \Rightarrow AB = 2AM$

$$AB = 2L\sin\theta = \frac{2LR}{\sqrt{L^2+R^2}}$$

(iii) ar(PAB)

Sol^n $ar(PAM) = \frac{1}{2} \times L\sin\theta \times L\cos\theta$

$$ar(PAB) = 2\,ar(PAM) = L^2\sin\theta\cdot\cos\theta$$

$$\Rightarrow ar(PAB) = \frac{L^3R}{L^2+R^2}$$

(iv) Angle b/w Pair of Tangents

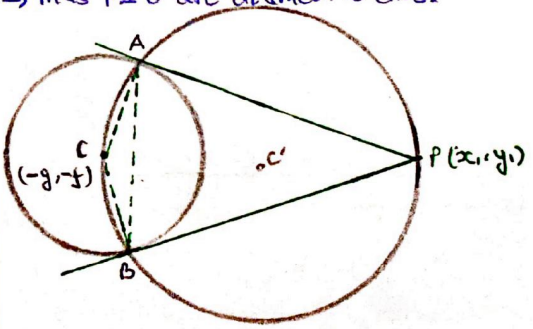
Sol^n $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2(R/L)}{1-(R/L)^2}$

$$\Rightarrow \tan 2\theta = \frac{2LR}{L^2-R^2}$$

(v) Eq^n of Circumcircle of △PAB

→ Circumcircle of △PAB will also inscribe □PACB as it is a cyclic quadrilateral.

→ Thus P & C are diametric ends.



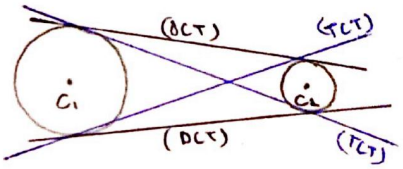
$$S_C: (x-x_1)(x+g)+(y-y_1)(y+f)=0$$

(Circumcircle's eqn)

Common Tangents:

Direct Common Tangent (DCT) (External)
⇓
Here, centre of circles lie on same side of tangents.

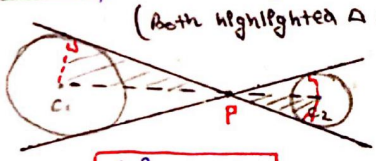
Transverse Common Tangent (TCT) (Internal)
⇓
Here, centre of circles lie on opp. sides of tangents.



3 Golden Points ⇒

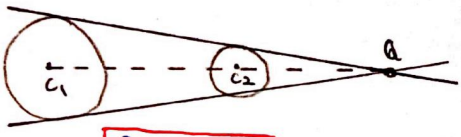
1) Both TCT intersect each other at line joining both centres & divide line C_1C_2 internally into ratio of their radii.

(Both highlighted △ are similar)



$$\frac{C_1P}{C_2P} = \frac{r_1}{r_2}$$

2) Both DCT intersect at line C_1C_2 and divide it into ratio of their radii externally.

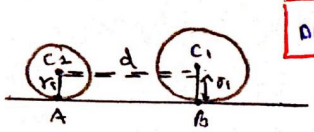


$$\frac{C_1Q}{C_2Q} = \frac{r_1}{r_2}$$

3) P & Q are harmonic conjugate of each other w.r.t C_1C_2 i.e. C_1P, C_1C_2 & C_1Q are in HP.

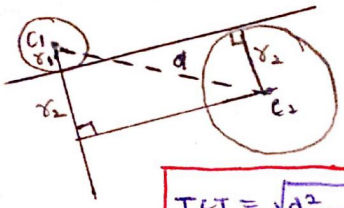
$$\frac{2}{C_1C_2} = \frac{1}{C_1P} + \frac{1}{C_1Q}$$

Length of DCT ⇒



$$DCT = \sqrt{d^2-(r_2-r_1)^2}$$

Length of TCT ⇒



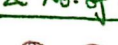
$$TCT = \sqrt{d^2 - (r_1 + r_2)^2}$$

⇒ Thus,

$$(\text{Length})_{TCT} < (\text{Length})_{DCT}$$

Position of Circles & No. of Common Tangents ⇒

1) $d > (r_1 + r_2)$



4 Common Tangents → 2 TCT, → 2 DCT

2) $d = (r_1 + r_2)$



3 Common Tangents → 2 DCT, → 1 TCT

3) $|r_2 - r_1| < d < (r_1 + r_2)$



2 Common Tangents → 2 DCT

4) $d = |r_1 - r_2|$



1 Common DCT

5) $d < |r_1 - r_2|$

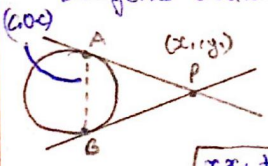


↳ No Common Tangent

Three Important Equations: (Valid for all conics)

(I) Chord of Contact ⇒

→ Chord joining Points of contacts of two tangents drawn from an ext. point $P(x_1, y_1)$



$$T = 0$$

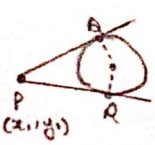
i.e.

$$xx_1 + yy_1 - a^2 = 0$$

OR

$$xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$$

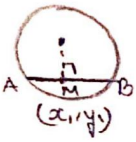
(II) Equation of Pair of Tangents ⇒



$$SS_1 = T^2$$

$$(x^2 + y^2 - a^2)(x_1^2 + y_1^2 - a^2) = (xx_1 + yy_1 - a^2)^2$$

(III) Eqn of chord with given mid-point ⇒



$$T = S_1$$

$$(xx_1 + yy_1 - a^2) = (x_1^2 + y_1^2 - a^2)$$

→ Of all the chords which passes through a given point M(a,b) inside the circle the shortest chord is one whose middle point is (a,b).

Family of Circles :

Type-01 ⇒

Eqn of the family of circles passing through the p.o.i of two two circles $S_1 = 0$ & $S_2 = 0$

$$S_1 + \lambda S_2 = 0 \quad (\lambda \neq -1)$$



Type-02 ⇒

Eqn of family of circles passing through P.O.I of a line $L = 0$ & circle $S = 0$ is

$$S + \lambda L = 0$$



Type-03 ⇒

Eqn of family of circles passing through two points (x_1, y_1) and (x_2, y_2).

→ We can write eqn by $S + \lambda L$



$$\Rightarrow (x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

(Eqn of Line)

Type-04 ⇒

Eqn of family of circle touching a line L at its fixed point (x_1, y_1)



$$(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$$

Angle of Intersection :

Draw tangents at P.O.I of two circles & find angle b/w them.

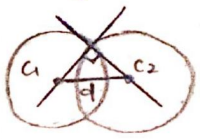


$$\cos\theta = \frac{d^2 - (r_1^2 + r_2^2)}{2 r_1 r_2}$$

d → Distance b/w centres

Condⁿ of Orthogonality ⇒

↳ When angle b/w circles $(\theta) = 90°$, then tangent at one circle passes through centre of another.

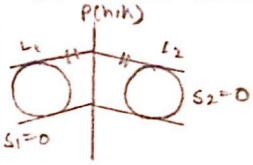


$$r_1^2 + r_2^2 = d^2$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

Radical Axis of a Circle Pair :

→ It is locus of points whose powers w.r.t two circles are equal.



Eqn of Radical Axis

$$S_1 - S_2 = 0$$

Important Points ⇒

1) If two circles intersect, then the radical ~~circles~~ axis is the common chord.

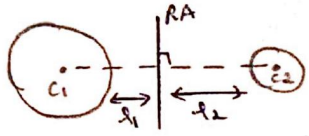


2) If two circles touch each other, then radical axis is common tangent



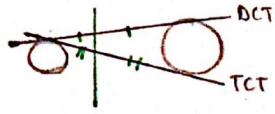


3) RA is always ⊥ar to the line joining centres.



4) RA is always closer to smaller circle, & bisects C_1C_2 if $r_1 = r_2$ only

5) RA bisects the common tangents.



6) Pairs of circles which do not have RA are concentric.

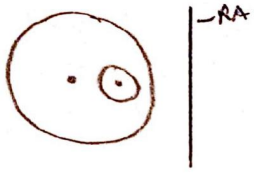
$S_1 : x^2 + y^2 + 2gx + 2fy + c = 0$

$S_2 : x^2 + y^2 + 2gx + 2fy + c = 0$



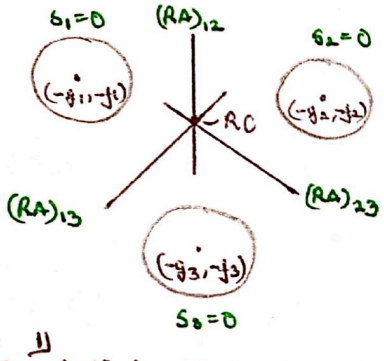
Circles-3

1] If one circle is contained in another circle then RA passes outside to both the circles.



#Radical Centre :

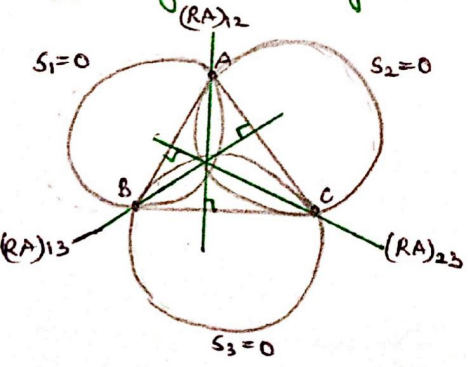
Point of intersection of radical axis of three circles (whose centres are non-collinear) taken two at a time, is Radical centre.



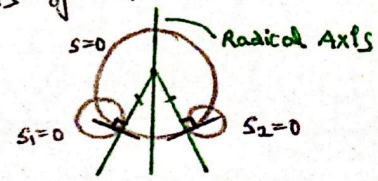
1]

NOTE : Radical centre of three circles described on the side of a Δ as diameter is the Orthocentre of that circle.

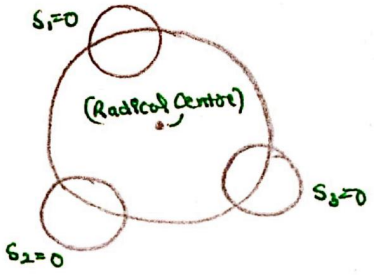
↳ Means corresponding RA would be along altitudes of the Δ.



2] Locus of centre of a variable circle S which cuts two fixed circles S_1 & S_2 orthogonally is the radical axis of S_1 & S_2.



3] Centre of a circle othrogonal to three circles S_1, S_2 & S_3 lies on their Radical centre.

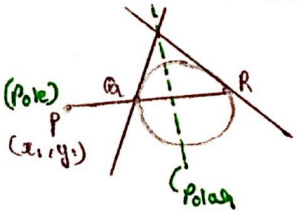


#Pole and Polar :

→ Through a point P in the plane of circle, draw a secant to the circle cutting at Q and R.

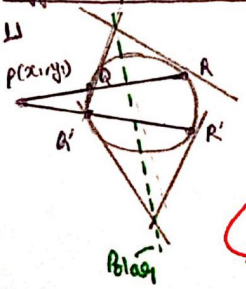
→ Then, subtent tangents at the Q points Q and R.

→ So the locus of P.O.I of tangents at Q & R is called 'Polar', and point P from which secant was drawn is called 'Pole of Polar'



Different positions of pole ⇒

1]

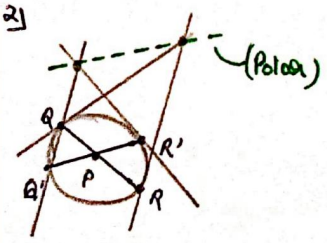


Outside the circle, polar eqn is given by

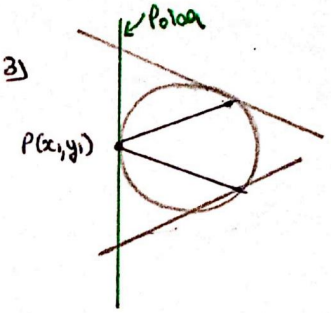
$T=0$ (w.r.t P)

↳ Acts as chord of contact

2]



3]



When pole lies on circumference of the circle, then polar eqn is

$T=0$

↳ Polar acts as a Tangent